Nilpotent orbits

Luiz Alberto

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Chapter 1

Kac - Moody algebras and its representations

1.1 Kac - Moody algebras

1.1.1 Main definitions

Key definitions that we need to remember:

- (Generalized Cartan matrix). A matrix $A = (a_{ij}) \in M_n(\mathbb{Z})$ is said to be a Generalized Cartan matrix if it satisfies:

$$\begin{cases} a_{ii} = 2, \ \forall i \geq 1 \\ a_{ij} \leq 0, \ \forall i \neq j \\ a_{ij} = 0 \iff a_{ji} = 0 \end{cases} \tag{1.1}$$

- (Realization of a matrix). If $A = (a_{ij}) \in M_n(\mathbb{Z})$ is a matrix of rank ℓ, then a realization of A is defined as a triple $(\mathfrak{h}, \Pi, \check{\Pi})$, where $\Pi = \{\alpha_1, \ldots, \alpha_n\} \subset \mathfrak{h}^*$ and $\check{\Pi} = \{\alpha_1^\vee, \ldots, \alpha_n^\vee\}$ satisfies:

$$\begin{cases} \Pi, \check{\Pi} \ \text{ are L.I sets} \\ \langle \alpha_i, \alpha_j^\vee \rangle = a_{ij} \end{cases} \tag{1.2}$$

  We claim that under this condition we have $\dim(\mathfrak{h}) \geq 2n - \ell$. Assume that $(\mathfrak{h}, \Pi, \check{\Pi})$ is a realization of A and write $\dim(\mathfrak{h}) = n + m$. So, we can extend $\check{\Pi}$ to a basis of $\mathfrak{h}$ by adding L.I vectors $v_1, \ldots, v_m$. Write $b_{ij} = \langle \alpha_i, v_j \rangle$, then since Π is a linearly independent set, the matrix $M = (A \ B)$, $B = (b_{ij}) \in M_{nm}(\mathbb{C})$ must be of rank n. However, by reordering the indexes, we can assume that $A = \left(\begin{smallmatrix} A_1 \\ A_2 \end{smallmatrix} \right), B = \left(\begin{smallmatrix} B_1 \\ B_2 \end{smallmatrix} \right)$, where $A_1 \in M_{\ell,n}, B_2 \in M_{n-\ell,m}$ and A_1 has full rank. Then we have

$$M = \begin{pmatrix} A_1 & B_1 \\ A_2 & B_2 \end{pmatrix}, \ \text{Rank}(M) = n. \tag{1.3}$$

Therefore $\mathrm{Rank}(B_2) = n - \ell \Rightarrow m \geq n - \ell$. This proves that $\dim(\mathfrak{h}) \geq 2n - \ell$. In general we will assume that a realization is minimum in the sense that $\dim(\mathfrak{h}) = 2n-\ell$.

- (Existence of realizations). Given a matrix $A = (a_{ij}) \in M_n(\mathbb{C})$, by reordering indexes we can write $A = \left(\begin{smallmatrix} A_1 \\ A_2 \end{smallmatrix}\right), \mathrm{Rank}(A_1) = \ell$. So the matrix

$$C = \begin{pmatrix} A_1 & 0 \\ A_2 & \mathrm{I}_{n-\ell,n-\ell} \end{pmatrix} \in M_{n,2n-\ell} \tag{1.4}$$

has full rank, so its rows are linearly independent. Then we let $\alpha_j^\vee = R_j(C)$, the rows of C and α_i the first n cordinate functions. This clearly gives a minimum realization of A.

- (Free Lie algebras). Let X be a set and $\mathbb{C}X$ the associated vector space of maps $\{X \to \mathbb{C} : \text{fin. support}\}$. We write $F(X) = T(\mathbb{C}X)$, which is an associative algebra. We also can view this as a Lie algbra with bracket $[a, b] = ab - ba$.

Definition 1.1.1. The free Lie algebra $FL(X)$ generated by X is the smallest Lie subalgebra of $F(X)$ containing X. Or equivalently, it's the intersection of all Lie subalgebrs of $F(X)$ containing X.

The imporantance of this definition lies in the following proposition.

Lemma 1.1.2. If $\theta : X \to \mathfrak{g}$ is a map from X to a Lie algebra $\mathfrak{g}$, then there exists an unique homomorphism $\phi : FL(X) \to \mathfrak{g}$ such that $\phi|_X = \theta$.

Proof: By the universal property of the tensor algebra, the map $\theta : X \to \mathfrak{g}$ induces a map of associative algebras $F(X) \to T(\mathfrak{g})$. So we have an induced map $\phi : F(X) \to \mathcal{U}(\mathfrak{g})$. We see that $\phi^{-1}(\mathfrak{g}) \subset F(X)$ is a Lie subalgebra which contains X, so $FL(X) \subset \phi^{-1}(\mathfrak{g})$ and we conclude that $\phi : FL(X) \to \mathfrak{g}$. It's clear that $\phi|_X = \theta$. The uniqueness follows from the fact that if ϕ' is another map $FL(X) \to \mathfrak{g}$ such that $\phi'|_X = \theta$, then the set of elements $a \in F(X)$ such that $\phi(a) = \phi'(a)$ is a Lie algebra containing X, so $\phi|_{FL(X)} = \phi'|_{FL(X)}$. $\square$

- (Pre Kac - Moody algebra). Let $A = (a_{ij}) \in M_n(\mathbb{C})$ and $(\mathfrak{h}, \Pi, \check{\Pi})$ be a realization of A. We let

$$X = \{f_i, e_i : i = 1, \ldots, n\} \cup \{\mathfrak{h}\}. \tag{1.5}$$

So we have the free Lie algebra $FL(X)$. We observe that $i : \mathfrak{h} \hookrightarrow FL(X)$ is just an inclusion of sets, not a vector space monomorphism. In order to make it a vector space monomorphism, we let

$$\mathfrak{h} \hookrightarrow \frac{FL(X)}{\langle i(\lambda x - \mu y) - \lambda i(x) + \mu i(y) \rangle}. \tag{1.6}$$

Now we add the relations (we will identify $\mathfrak{h}$ and $i(\mathfrak{h})$):

$$\begin{cases} [h, e_i] = \langle \alpha_i, h\rangle e_i \\ [h, f_i] = -\langle \alpha_i, h\rangle f_i \\ [e_i, f_j] = \delta_{ij}\alpha_i^\vee \end{cases} \tag{1.7}$$

for all $h \in \mathfrak{h}$. Call all these relations by R, we will write

$$\widetilde{\mathfrak{g}}(A) = \frac{FL(X)}{R}. \tag{1.8}$$

- (A tensor algebra representation for $\widetilde{\mathfrak{g}}(A)$). Let $V = \bigoplus_{i=1}^n \mathbb{C}v_i$, there is a natural construction of a representation of $\widetilde{\mathfrak{g}}(A)$ on $T(V)$ for each $\lambda \in \mathfrak{h}^*$. The defining relations are as follows:

$$\begin{cases} f_i(a) = v_i \otimes a, \;\; a \in T^s(V) \\ h(1) = \langle \lambda, h\rangle 1 \\ e_i(1) = 0 \end{cases} \tag{1.9}$$

Then we use the “creation” operator f_i to deduce the action of h and e_i on $T(V)$ inductively.

It's important to note that the defined action does not looks like a derivation ! That is, we dont have in general the relation:

$$X(a \otimes b) = (Xa) \otimes b + a \otimes (Xb), \;\; a \in T^s(V), b \in T^r(V). \tag{1.10}$$

For example, we see that

$$\begin{aligned} h(v_i) = hf_i(1) &= f_i h(1) + [h, f_i](1) \\ &= \langle \lambda, h\rangle f_i(1) - \langle \alpha_i, h\rangle f_i(1) \\ &= \langle \lambda - \alpha_i, h\rangle v_i. \end{aligned} \tag{1.11}$$

Furthermore

$$\begin{aligned} h(v_i \otimes a) &= f_i h(a) + [h, f_i](a) \\ &= -\langle \alpha_i, h\rangle f_i(a) + v_i \otimes h(a), \;\; a \in T^s(V). \end{aligned} \tag{1.12}$$

In general we see that:

$$\begin{cases} e_i : T^s(V) \to T^{s-1}(V) : v_j \otimes a \mapsto \delta_{ij}\alpha_j^\vee(a) + v_j \otimes e_i(a) \\ f_i : T^s(V) \to T^{s+1}(V) : a \mapsto v_i \otimes a \\ h : T^s(V) \to T^s(V) : v_j \otimes a \mapsto -\langle \alpha_j, h\rangle v_j \otimes a + v_j \otimes h(a), \;\; h \in \mathfrak{h}. \end{cases} \tag{1.13}$$

In addition we see that $T(V)$ is a weight representation:

$$\begin{aligned} h \cdot v_{i_1} \otimes \cdots \otimes v_{i_k} &= \langle \lambda - \sum_{j=1}^{k} \alpha_{i_j}, h \rangle v_{i_1} \otimes \cdots \otimes v_{i_k} \\ \Rightarrow T(V) &= \bigoplus_{k_1, \ldots, k_n \geq 0} V_{\lambda - \Sigma_{j=1}^{n} k_j \alpha_j}. \end{aligned} \tag{1.14}$$

Using this family of representations we deduce the next corollary.

Corollary 1.1.3. We have $\widetilde{\mathfrak{g}}(A) = \widetilde{\mathfrak{n}}_- \oplus \mathfrak{h} \oplus \widetilde{\mathfrak{n}}_+$, where $\widetilde{\mathfrak{n}}_\pm$ are the subalgebras generated by the e's and f's respectively.

Proof: It's easy to see that $\widetilde{\mathfrak{g}}(A) = \widetilde{\mathfrak{n}}_- + \mathfrak{h} + \widetilde{\mathfrak{n}}_+$, now let $n_- + h + n_+ = 0 \in \widetilde{\mathfrak{g}}(A)$ and $\lambda \in \mathfrak{h}^*$. Using the representation constructed above we see that $n_-(1) + \langle \lambda, h \rangle 1 = 0$. This last expression is true for all $\lambda \in \mathfrak{h}^*$, then $h = 0$. Now it remains to prove that

$$\theta : \widetilde{\mathfrak{n}}_- \to T(V) : n_- \mapsto n_-(1) \tag{1.15}$$

is injective. To see this we observe that the above map is just the restriction of the $(T(V), \lambda)$ restricted to $\widetilde{\mathfrak{n}}_-$, so it's a Lie algebra homomorphism. Its image contains $\{v_1, \ldots, v_n\}$ so

$$\theta : \widetilde{\mathfrak{n}}_- \to FL(V). \tag{1.16}$$

However, this map has an inverse given by $v_i \mapsto f_i$. Then we conclude that θ is injective. □

- (More notation). We have the set of notations.

$$\begin{cases} Q = \sum_i \mathbb{Z}\alpha_i, Q_+ = \sum_i \mathbb{Z}_{\geq 0}\alpha_i,\ Q_- = -Q_+ \\ \mathfrak{g}_\alpha = \{x \in \widetilde{\mathfrak{g}}(A) : [h, x] = \langle \alpha, h \rangle x\} \\ \text{(These implies:) } \widetilde{\mathfrak{g}}(A) = (\bigoplus_{\alpha \in Q_- \setminus 0} \mathfrak{g}_\alpha) \oplus \mathfrak{h} \oplus (\bigoplus_{\alpha \in Q_+ \setminus 0})\mathfrak{g}_\alpha \\ \mathfrak{g}_j(s) = \bigoplus_\alpha \mathfrak{g}_\alpha,\ \alpha = \sum_i k_i \alpha_i,\ \sum_i k_i s_i = j \\ \mathfrak{g}_j = \mathfrak{g}_j(1, \ldots, 1) = \bigoplus_{\alpha\,:\,\mathrm{ht}(\alpha) = j} \mathfrak{g}_\alpha \end{cases} \tag{1.17}$$

- (The Kac - Moody algebra). We define

$$\mathfrak{g}(A) = \frac{\widetilde{\mathfrak{g}}(A)}{I} \tag{1.18}$$

where I is the maximal ideal which intersects $\mathfrak{h}$ trivially. This maximal ideal exists because the general result that

$$V = \bigoplus_{\lambda} V_\lambda \Rightarrow W = \bigoplus_{\lambda} (W \cap V_\lambda) \tag{1.19}$$

for any weight representation V of a Lie algebra and a submodule $W \subset V$. Therefore, the sum of ideals intersecting $\mathfrak{h}$ trivially, is an ideal intersecting $\mathfrak{h}$ trivially.

More notation:

$$\begin{cases} \mathfrak{g}_\alpha = \{x \in \mathfrak{g}(A) : [h, x] = \langle \alpha, h \rangle x, \ \forall h \in \mathfrak{h}\} \\ \text{mult}(\alpha) = \dim(\mathfrak{g}_\alpha) \\ \Delta_+ = \{\alpha \in Q_+ : \text{mult}(\alpha) \neq 0\}, \ \Delta_- = -\Delta_+ \\ \text{(The root system:) } \Delta = \Delta_+ \cup \Delta_- \end{cases} \tag{1.20}$$

1.1.2 Invariant bilinear form

A generalized Cartan matrix A is said to be symmetrizable if there exist a decomposition

$$A = DB, \ D = \text{diag}(\epsilon_1, \dots, \epsilon_n), B = (b_{ij}) \text{ symmetric.} \tag{1.21}$$

We also impose the condition on D being invertible, i.e., $\epsilon_i \neq 0 \ \forall i$.

Let $\mathfrak{g} = \mathfrak{g}(A)$ be a Kac - Moody algebra, with A being a symmetrizable generalized Cartan matrix. We define a bilinear form $(\cdot|\cdot) : \mathfrak{h} \times \mathfrak{h} \to \mathbb{C}$ as follows:

$$\begin{cases} \mathfrak{h} = \mathfrak{h}' + \mathfrak{h}'', \ \mathfrak{h}' = \sum_i \mathbb{C}\alpha_i^\vee \\ (\alpha_i^\vee, |h) = \langle \alpha_i, h \rangle \epsilon_i, \ \forall h \in \mathfrak{h} \\ (h'|h'') = 0, \ \forall h', h'' \in \mathfrak{h}'' \end{cases} \tag{1.22}$$

Theorem 1.1.4. There exist an unique symmetric bilinear form on $\mathfrak{g}(A)$ that's invariant, nondegenerate on $\mathfrak{h}$ and satisfying relations (1.22).

Proof: If $(\cdot|\cdot)$ satisfies (1.22), then it's nondegenerate on $\mathfrak{h}$. In fact, if $(\sum_i c_i \alpha_i^\vee | h) = 0, \ \forall h \in \mathfrak{h}$, then $\langle \sum_i c_i \epsilon_i \alpha_i, h \rangle = 0, \ \forall h \in \mathfrak{h}$. Therefore

$$\sum_i c_i \epsilon_i \alpha_i = 0 \Rightarrow c_i = 0, \ \forall i. \tag{1.23}$$

Now, by invariance we see that this form must satisfies $(x|y) = 0$ whenever $(x, y) \in \mathfrak{g}_\alpha \times \mathfrak{g}_\beta, \alpha + \beta \neq 0$. In fact, for any $h \in \mathfrak{h}$, we have

$$\langle \alpha, h \rangle (x|y) = ([h, x]|y) = -(x|[h, y]) = -\langle \beta, h \rangle (x|y) \tag{1.24}$$

and $\langle \alpha+\beta, h\rangle (x|y) = 0$. Since $\alpha+\beta \neq 0$, we see that $(x|y) = 0$. From this we also deduce that $(\cdot|\cdot)|_{\mathfrak{g}_\alpha \times \mathfrak{g}_{-\alpha}}$ is nondegenerate, otherwise $(\cdot|\cdot)$ will be degenerate, which means that $\mathrm{Ker}(\cdot|\cdot)$ would be a nonzero ideal that intersects $\mathfrak{h}$ trivially, which is a contradiction.

We also can use invariance to find a formula for $(\cdot|\cdot)$ on $\mathfrak{g}(N) = \bigoplus_{j=-N}^{N} \mathfrak{g}_j$. First, on $\mathfrak{g}(1) = \sum_i \mathbb{C} f_i + \mathfrak{h} + \sum_i \mathbb{C} e_i$, we already know that we must have

$$\begin{cases} (\mathfrak{g}_{\pm 1}, \mathfrak{g}_{\pm 1}) = 0 \\ (\mathfrak{g}_{\pm 1}, \mathfrak{h}) = 0 \end{cases} \tag{1.25}$$

It still remains to calculate $(e_i|f_j)$. By invariance, we must have

$$\langle \alpha_i, h\rangle (e_i|f_j) = (e_i|[f_j, h]) = ([e_i, f_j]|h) = \delta_{ij}(\alpha_i^\vee|h). \tag{1.26}$$

So, for $h = \alpha_j^\vee$ we deduce the relation $(e_i|f_j) = \delta_{ij}\epsilon_i$. Now, assume that we have defined $(\cdot|\cdot)$ on $\mathfrak{g}(N-1)$. To extend this form to $\mathfrak{g}(N)$ we need to define it on $\mathfrak{g}_N \times \mathfrak{g}_{-N}$. So let $(x, y) \in \mathfrak{g}_N \times \mathfrak{g}_{-N}$ and write $x = \sum_i [u_i, v_i]$ where $\deg(u_i) + \deg(v_i) = N, u_i, v_i \in \mathfrak{g}(N-1)$. Although $([u_i, v_i], y)$ is not yet defined, we can use invariance to write

$$(x|y) = \sum_i (u_i|[v_i, y]) \tag{1.27}$$

because $\deg([v_i, y]) > -N$. That turns out to be a definition for the extension of the bilinear form from $\mathfrak{g}(N-1)$ to $\mathfrak{g}(N)$. $\square$

Since $(\cdot|\cdot)$ is nondegenerate, we have an induced map $\nu : \mathfrak{h} \to \mathfrak{h}^*$. It's clear from equation (1.22) that $\nu(\alpha_i^\vee) = \epsilon_i \alpha_i$. Therefore

$$\begin{aligned} (\alpha_i|\alpha_i) &= \epsilon_i^{-2}(\alpha_i^\vee|\alpha_i^\vee) \\ &= \epsilon_i^{-2}\epsilon_i\langle \alpha_i^\vee, \alpha_i\rangle \\ &= \epsilon_i^{-1} 2 \\ &\Rightarrow \epsilon_i = \frac{2}{(\alpha_i|\alpha_i)}. \end{aligned} \tag{1.28}$$

So we conclude that $\alpha_i^\vee = \frac{2}{(\alpha_i|\alpha_i)}\nu^{-1}(\alpha_i)$ and $A_{ij} = 2\frac{(\alpha_j|\alpha_i)}{(\alpha_j|\alpha_j)}$.

1.1.3 Weyl group of a Kac - Moody algebra

A quick sketch for this subsection:

$$\begin{tikzcd}
& \text{Representation theory of } \mathfrak{sl}_2\mathbb{C} \arrow[dl] \arrow[dd] \\
\mathrm{ad}(e_i),\ \mathrm{ad}(f_i) \text{ are nilpotent operators on } \mathfrak{g}(A) \arrow[d] \arrow[dr, dashed] & \\
W = \left\langle r_i = \exp(f_i)\exp(-e_i)\exp(f_i) \right\rangle \arrow[r, dashed] & \dim(V_\lambda) = \dim(V_{r_i(\lambda)})
\end{tikzcd}$$

Lemma 1.1.5. On a Kac - Moody algebra $\mathfrak{g}(A)$, the operators $ad(e_i)$ and $\mathrm{ad}(f_i)$ are locally nilpotent.

Proof: The proof include several steps:

- (A little lemma). If $a \in \mathfrak{n}_+$ is such that $[f_i, a] = 0,\ \forall i$, then $a = 0$. In fact, if $a \neq 0$, then
$$\sum_{i,j\geq 0} \mathrm{ad}(\mathfrak{g}_1)^i \mathrm{ad}(\mathfrak{h})^j a \tag{1.29}$$
would be a nonzero ideal intersecting $\mathfrak{h}$ trivially, which is a contradiction.

- (Representation theory of $\mathfrak{sl}_2$). Recall that if V is a $\mathfrak{sl}_2\mathbb{C} = \mathbb{C}\langle Y, H, X\rangle$ module and $v \in V$ is a vector satisfying:
$$\begin{cases} H(v) = \lambda v \\ X(v) = 0 \end{cases} \tag{1.30}$$
Then $\lambda = n \in \mathbb{Z}_{\geq 0}$ and $XY^k(v) = k(n-k+1)Y^{k-1}(v)$. In fact, $W = \langle Y^k(v) : k \geq 0\rangle$ is an irreducible $\mathfrak{sl}_2\mathbb{C}-$module.

- (Serre relations). We have
$$\mathrm{ad}(e_i)^{1-a_{ij}} e_j = 0 = \mathrm{ad}(f_i)^{1-a_{ij}} f_j \text{ for } i \neq j. \tag{1.31}$$
We will shot that $\mathrm{ad}(f_i)^{1-a_{ij}} f_j = 0$ by proving that it commutes with all the e's. If we write $v = f_j$ and think $\mathfrak{g}(A)$ as a representation under the adjoint action of $\mathfrak{sl}_2\mathbb{C} = \mathbb{C}\langle f_i, \alpha_i^\vee, e_i\rangle$, then
$$e_i(v) = 0,\ \alpha_i^\vee(v) = -a_{ij} v. \tag{1.32}$$
Then we see that
$$\begin{aligned} [e_i, \mathrm{ad}(f_i)^{1-a_{ij}} f_j] &= e_i \cdot f_i^{1-a_{ij}}(v) \\ &= (1-a_{ij})(-a_{ij} - (1-a_{ij}) + 1) f_i^{-a_{ij}}(v) \\ &= 0. \end{aligned} \tag{1.33}$$
Moreover $[e_k, \mathrm{ad}(f_i)^{1-a_{ij}} f_j] = 0, \forall k \neq i, j$. Finally, we have
$$\begin{aligned} [e_j, \mathrm{ad}(f_i)^{1-a_{ij}} f_j] &= \mathrm{ad}(f_i)^{1-a_{ij}} \alpha_j^\vee \\ &= a_{ij} \mathrm{ad}(f_i)^{-a_{ij}}(f_i) \\ &= 0, \text{ if } a_{ij} < 0. \end{aligned} \tag{1.34}$$
For $a_{ij} = 0$ we still have $[e_j, [f_i, f_j]] = [f_i, \alpha_j^\vee] = a_{ij}\alpha_j^\vee = 0$.

- (Conclusion). Said this, we know that for each j, there exist $N_j \geq 1$ such that $\mathrm{ad}(f_i)^{N_j} f_j = 0$. The same is true for the other generators of $\mathfrak{g}(A)$. Finally, an induction on the length of a Lie word on f's and e's proves that $\mathrm{ad}(f_i)$ is locally nilpotent on $\mathfrak{g}(A)$.

Definition 1.1.6. We define the Weyl group of $\mathfrak{g}(A)$ by

$$W = \left\langle \theta_i^{\mathrm{ad}} = e^{\mathrm{ad}(e_i)} e^{-\mathrm{ad}(f_i)} e^{\mathrm{ad}(e_i)}\big|_{\mathfrak{h}} \right\rangle \subset GL(\mathfrak{h}). \tag{1.35}$$

Lemma 1.1.7. We have

$$\theta_i(h) = h - \langle \alpha_i, h \rangle \alpha_i^\vee. \tag{1.36}$$

Proof: It's clear that if $\langle \alpha_i, h \rangle = 0$, then $\exp(\mathrm{ad}(e_i))h = h = \exp(\mathrm{ad}(f_i))h$. Therefore, in this case we have $\theta_i(h) = h$. So it remains to show that $\theta_i(\alpha_i^\vee) = -\alpha_i^\vee$. To see this we use the standard representation of $\mathfrak{sl}_2\mathbb{C}$ to write

$$e_i = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, f_i = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \alpha_i^\vee = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{1.37}$$

So, we see that

$$\exp(e_i) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \exp(-f_i) = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}. \tag{1.38}$$

Therefore

$$\begin{aligned} \theta_i^{\mathrm{ad}}(\alpha_i^\vee) &= \mathrm{Ad}_{\exp(e_i)} \mathrm{Ad}_{\exp(-f_i)} \mathrm{Ad}_{\exp(e_i)} \alpha_i^\vee \\ &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= -\alpha_i^\vee. \end{aligned} \tag{1.39}$$

□

Taking the contragradient representation of W on $\mathfrak{h}^*$ and using the formula (1.36), we arrive at an equivalent definition for the Weyl group.

Definition 1.1.8. We define the Weyl group of $\mathfrak{g}(A)$ by the group generated by reflections defined by

$$r_i(\lambda) = \lambda - \langle \lambda, \alpha_i^\vee \rangle \alpha_i. \tag{1.40}$$

Corollary 1.1.9. Assume that V is an integrable $\mathfrak{g}(A)-$module, i.e., V is a weight representation

$$V = \bigoplus_{\lambda\in\mathfrak{h}^*} V_\lambda \tag{1.41}$$

and each e_i, f_i acts on V as nilpotent operators. Then $\theta_i(V_\lambda) = V_{r_i(\lambda)}$, where we let

$$\theta_i = \exp(e_i)\exp(-f_i)\exp(e_i) \in \mathrm{End}(V). \tag{1.42}$$

In particular $\dim(V_{w(\lambda)}) = \dim(V_\lambda)$, $\forall w \in W$.

Proof: First, remember some formulas. Let $X, T \in \mathrm{End}(V)$ be nilpotent operators on V. Then we have

$$\begin{aligned}
\exp(T)X\exp(-T) &= \sum_{i,j\geq 0} \frac{(-1)^j}{i!j!} T^i X T^j \\
&= \sum_{m\geq 0} \frac{1}{m!} \sum_{j=0}^{m} \binom{m}{j} (-1)^j T^{m-j} X T^j \\
&= \sum_{m\geq 0} \frac{1}{m!} (L_T - R_T)^m X \\
&= \sum_{m\geq 0} \frac{1}{m!} \mathrm{ad}(T)^m X \\
&= \exp(\mathrm{ad}(T))X.
\end{aligned} \tag{1.43}$$

If $\theta_i = \exp(e_i)\exp(-f_i)\exp(e_i)$, then by the previous equation we see that

$$\begin{aligned}
\theta_i \cdot \alpha_i^\vee \cdot \theta_i^{-1} &= \exp(\mathrm{ad}(e_i))\exp(\mathrm{ad}(-f_i))\exp(\mathrm{ad}(e_i))\alpha_i^\vee \\
&= \theta_i^{\mathrm{ad}}(\alpha_i^\vee) \\
&= -\alpha_i^\vee.
\end{aligned} \tag{1.44}$$

Let $v \in V_\lambda$, then it's clear that $h \cdot \theta_i(v) = \langle \lambda, h\rangle \theta_i(v)$ because h commute with e_i and f_i. However, by the previous calculation we see that we also have $\alpha_i^\vee \theta_i(v) = -\theta_i \alpha_i^\vee(v) = -\langle \lambda, \alpha_i^\vee\rangle \theta_i(v)$. Therefore

$$\begin{aligned}
(h - \frac{1}{2}\langle \lambda, h\rangle \alpha_i^\vee) \cdot \theta_i(v) = \langle \lambda, h - \frac{1}{2}\langle \lambda, h\rangle \alpha_i^\vee\rangle \theta_i(v) \Rightarrow \\
h \cdot \theta_i(v) = \langle \lambda, h - \langle \alpha_i, h\rangle \alpha_i^\vee\rangle \theta_i(v) \\
= \langle \lambda, \theta_i^{\mathrm{ad}}(h)\rangle \theta_i(v) \\
= \langle r_i(\lambda), h\rangle \theta_i(v).
\end{aligned} \tag{1.45}$$

□

Lemma 1.1.10. The invariant bilinear form defined on $\mathfrak{g}(A)$ is preserved under the action of the Weyl group.

Proof: Let $a, b, x \in \mathfrak{g}(A)$, then we deduce (from invariance):

$$(\mathrm{ad}(x)^i a|b) = (-1)^i (a|\mathrm{ad}(x)^i b). \tag{1.46}$$

Therefore

$$(\exp(ad(x))a|\exp(\mathrm{ad}(x))b) = (a|\exp(-\mathrm{ad}(x))\exp(\mathrm{ad}(x))b) = (a|b). \tag{1.47}$$

So we see that $(\theta_i^{\mathrm{ad}}(a)|\theta_i^{\mathrm{ad}}(b)) = (a|b)$. □

1.1.4 Affine Lie algebras

- (Affine Cartan matrix). An affine GCM is a GCM satisfying:

$$\begin{aligned} &(1)\ \mathrm{Corank}(A) = 1 \\ &(2)\ Au = 0 \text{ for some } u > 0 \\ &(3)\ Au \geq 0 \Rightarrow Au = 0 \end{aligned} \tag{1.48}$$

  For example, lets classify 2×2 generalized Cartan matrices of affine type. If $A = \left(\begin{smallmatrix} 2 & a \\ b & 2 \end{smallmatrix}\right)$ is affine, then by the first condition it has rank 1, so $\det(A) = 4 - ab = 0$ and since $a, b \in \mathbb{Z}_{\leq 0}$ we see that $a \in \{-1, -2, -4\}$. We see two possibilities (take into acount the transposes):

$$\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix} \text{ and } \begin{pmatrix} 2 & -1 \\ -4 & 2 \end{pmatrix}. \tag{1.49}$$

  We still need to see that the other two conditions are satisfied. For the second one we take $u = (1, 1)$ and $u = (1, 2)$ for the two matrices respectively. The third condition is clearly satisfies for both matrices.

- (Realization by loop algebras extensions). If $\mathfrak{g}$ is a finite dimensional Lie algebra with an invariant bilinear form $(\cdot, \cdot)$, we have affinization given by

$$\widehat{g} = \mathfrak{g}[t^{\pm}] + \mathbb{C}K,\ [xt^m, yt^n] = [x, y]t^{m+n} + m\delta_{m,-n}(x, y)K. \tag{1.50}$$

  The Kac - Moody affinization, is an extension of $\widehat{\mathfrak{g}}$ by a derivation. We have

$$\widetilde{g} = \widehat{g} + \mathbb{C}d,\ [d, K] = 0, [d, xt^m] = mxt^m. \tag{1.51}$$

  In the Kac - Moody affinization, the vector space $\widetilde{\mathfrak{h}} = \mathfrak{h} + \mathbb{C}K + \mathbb{C}d$ plays the role of Cartan subalgebra. We will show that $\widetilde{g}$ is an affine Kac moody algebra.

- (Explicty realization for affine Cartan matrices). Let $\mathfrak{g}$ be a simple finite dimensional Lie algebra and choose a Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$ and a set of positive roots. Then we have a triangular decomposition

$$\mathfrak{g} = \bigoplus_{\alpha\in\Delta_+} \mathfrak{g}_{-\alpha} + \mathfrak{h} + \bigoplus_{\alpha\in\Delta_+} \mathfrak{g}_{\alpha}. \tag{1.52}$$

Let $\Pi = \{\alpha_1, \ldots, \alpha_\ell\}$ be the set of simple roots (so $\mathrm{Rank}(\mathfrak{g}) = \ell$) and choose $(E_i, F_i) \in \mathfrak{g}_{\alpha_i} \times \mathfrak{g}_{-\alpha_i}$ in such a way that $\langle F_i, H_i, E_i\rangle$, where $H_i = [E_i, F_i]$, is a $\mathfrak{sl}_2-$triple. So $H_i = \alpha_i^\vee$, where we define in general

$$\begin{aligned} &\nu : \mathfrak{h} \overset{\cong}{\to} \mathfrak{h}^* : H \mapsto K(H,\ \cdot\ ), \\ &\alpha^\vee = \frac{2}{(\alpha,\alpha)}\nu^{-1}(\alpha),\ \alpha \in \Delta. \end{aligned} \tag{1.53}$$

Let $\theta \in \Delta_+$ be the (unique) maximal root of maximal heith: Remember that $\Delta_+ \subset \mathbb{Z}_{\geq 0}\Pi$ and $\mathrm{ht}(\sum_i k_i\alpha_i) = \sum_i k_i,\ \sum_i k_i\alpha_i \in \Delta_+$. For example, in $\mathfrak{g} = \mathfrak{sl}_{n+1}\mathbb{C}$ we have a triangular decomposition given by

$$\begin{cases} \mathfrak{g} = \bigoplus_{i>j} \mathfrak{g}_{L_i - L_j} + \mathfrak{h} + \bigoplus_{i<j} \mathfrak{g}_{L_i-L_j}, \\ \mathfrak{h} = \text{ traceless diagonal matrices}, \\ \mathfrak{g}_{L_i-L_j} = \mathbb{C}E_{ij},\ i \neq j \\ \Pi = \{\alpha_1, \ldots, \alpha_n\},\ \alpha_i = L_i - L_{i+1} \\ \theta = \sum_{i=1}^n \alpha_i = L_1 - L_{n+1} \\ K(\cdot,\cdot) = 2n\mathrm{Tr}(\cdot\cdot). \end{cases} \tag{1.54}$$

In particular, since $\langle E_{n+1,1}, H_{1,n+1}, E_{1,n+1}\rangle,\ H_{1,n+1} = E_{11} - E_{n+1,n+1}$, we have $\theta^\vee = H_{1,n+1}$.

So given a simple finite dimensional Lie algebra, and $\widetilde{\mathfrak{g}}$ its Kac - Moody affinization, we define

$$\begin{aligned} &\Pi^\vee = \{\underbrace{\tfrac{2}{(\theta,\theta)}K - \theta^\vee \otimes 1}_{\alpha_0^\vee}, \alpha_i^\vee : i = 1, \ldots, \ell\} \text{ where } \alpha_i^\vee = H_i \otimes 1\ (i > 0), \\ &\Pi = \{\underbrace{\delta - \theta}_{\alpha_0}, \alpha_i : i = 1, \ldots, \ell\}. \end{aligned} \tag{1.55}$$

Here we observe that we extend $\mathfrak{h}^* \hookrightarrow \widetilde{\mathfrak{h}}^*$ by putting $\alpha|_{\mathbb{C}K+\mathbb{C}d} = 0$ for $\alpha \in \mathfrak{h}^*$. Furthermore we define $\delta \in \widetilde{\mathfrak{h}}^*$ by $\delta(d) = 1, \delta|_{\mathfrak{h}+\mathbb{C}K} = 0$.

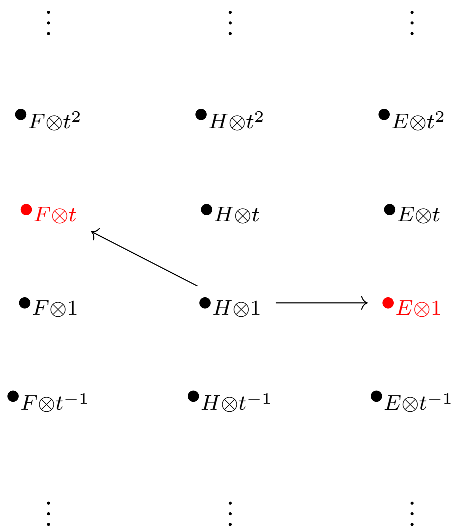
Define $A = (\langle \alpha_i, \alpha_j^\vee \rangle)_{i,j=0}^{\ell}$, we claim that it's an affine GCM (TODO: proof). For example, for $\mathfrak{g} = \mathfrak{sl}_{n+1}\mathbb{C}$ we have the induced affine GCM:

$$\begin{pmatrix} 2 & -1 & 0 & \cdots & 0 & -1 \\ -1 & 2 & -1 & \cdots & 0 & 0 \\ 0 & -1 & 2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 2 & -1 \\ -1 & 0 & 0 & \cdots & -1 & 2 \end{pmatrix} \tag{1.56}$$

Before we prove that $\widetilde{g}$ is the affine Kac - Moody algebra associated to the affine GCM A, we need to define a set of generators for it. We let

$$\begin{cases} e_0 = E_{-\theta} \otimes t, \ f_0 = E_\theta \otimes t^{-1} \\ e_i = E_i \otimes 1, \ f_i = F_i \otimes 1 \end{cases} \tag{1.57}$$

To understand this convention, we can look to the root system of $\widetilde{\mathfrak{sl}_2}$:



We see in general that

$$\begin{aligned} \widetilde{\Delta} &= \{\alpha + j\delta : \alpha \in \Delta, j \in \mathbb{Z}\} \cup \{j\delta : j \neq 0\} \\ \widetilde{g}_{\alpha+j\delta} &= \mathbb{C}E_\alpha \otimes t^j, \widetilde{g}_{j\delta} = \mathfrak{h}t^j. \end{aligned} \tag{1.58}$$

- (Main theorem). Before we prove that $\widetilde{\mathfrak{g}}$ is an Affine Kac - Moody algebra, we need a lemma.

Lemma 1.1.11. Let $\mathfrak{g}$ and suppose that:

- $\mathfrak{h} \subset \mathfrak{g}$ is a commutative subalgebra
- $\mathfrak{g} = \mathbb{C}\langle e_i, f_i : (i = 1, \dots, n), \mathfrak{h}\rangle$
- There exist $\Pi = \{\alpha_1, \dots, \alpha_n\} \subset \mathfrak{h}^*$ and $\Pi^\vee = \{\alpha_1^\vee, \dots, \alpha_n^\vee\} \subset \mathfrak{h}$ (both L.I sets) such that

$$\begin{aligned} [h, e_i] &= \langle \alpha_i, h\rangle e_i \\ [h, f_i] &= -\langle \alpha_i, h\rangle f_i \\ [e_i, f_i] &= \delta_{ij}\alpha_i^\vee \end{aligned} \tag{1.59}$$

  for all $h \in \mathfrak{h}$.
- Any nonzero ideal of $\mathfrak{g}$ intersects $\mathfrak{h}$ trivially

Then $(\mathfrak{g}, \mathfrak{h}, \Pi, \Pi^\vee)$ is a realization of $A = (\langle \alpha_i, \alpha_j^\vee\rangle)_{i,j=1}^n$.

Proof: Since $\mathfrak{g}$ is generated by $\mathfrak{h}$ and e_i, f_i $(i = 1, \dots, n)$ we have a surjective map

$$\varphi : \widetilde{\mathfrak{g}}(A) \twoheadrightarrow \mathfrak{g}. \tag{1.60}$$

Now $I = \mathrm{Ker}(\varphi) \subset \widetilde{\mathfrak{g}}(A)$ is an ideal. Since $\varphi|_{\mathfrak{h}} = \mathbb{1}_{\mathfrak{h}}$ by construction, we see that $I \cap \mathfrak{h} = (0)$. Moreover, if $J \subset \widetilde{\mathfrak{g}}(A)$ intersecting $\mathfrak{h}$ trivially, then $\varphi(J) \subset \mathfrak{g}$ is an ideal intersecting $\mathfrak{h}$ trivially, so $\varphi(J) = 0$ and $J \subset \mathrm{Ker}(\varphi)$. Therefore $\mathrm{Ker}(\varphi)$ is the maximal ideal of $\widetilde{\mathfrak{g}}(A)$ intersecting $\mathfrak{h}$ trivially and we have an induced isomorphism $\mathfrak{g}(A) \xrightarrow{\cong} \mathfrak{g}$. $\square$

Theorem 1.1.12. Let $\mathfrak{g}$ be a finite dimensional simple Lie algebra. Choose a Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$ and the induced triangular decomposition

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha\in\Delta} \mathfrak{g}_\alpha. \tag{1.61}$$

For a choice of positive roots we have a set of simple roots $\{\alpha_1, \dots, \alpha_\ell\}$ and the associated set of coroots $\{H_1, \dots, H_\ell\}$ defined by equation (1.53). Let θ be the (unique) root of maximal heith, then we define

$$\alpha_0 = \delta - \theta,\ \alpha_0^\vee = \frac{2}{(\theta,\theta)}K - \theta^\vee \otimes 1 \tag{1.62}$$

where $\delta \in \widetilde{\mathfrak{h}}^*, \widetilde{\mathfrak{h}} = \mathfrak{h} + \mathbb{C}K + \mathbb{C}d, \delta|_{\mathfrak{h}+\mathbb{C}K} = 0, \delta(d) = 1$. Furthermore, let $\Pi = \{\alpha_0, \dots, \alpha_\ell\}$ and $\Pi^\vee = \{\alpha_0^\vee, \dots, \alpha_\ell^\vee\}$, we claim that the quadruple $(\widetilde{\mathfrak{g}}, \widetilde{\mathfrak{h}}, \Pi, \Pi^\vee)$ is a realization of the affine GCM $(\langle \alpha_i, \alpha_j^\vee\rangle)_{i,j=0}^\ell$.

Proof: This proof have three parts:

– ($\widetilde{\mathfrak{g}}$ is generated by the e_i, f_i $(i = 0, \dots, \ell)$). Remember that we define

$$e_0 = E_{-\theta} \otimes t, f_0 = E_\theta \otimes t^{-1}, e_i = E_i \otimes 1, f_i = F_i \otimes 1 (i = 1, \dots, \ell) \tag{1.63}$$

where $E_i = E_{\alpha_i}, F_i = E_{-\alpha_i}$ are such that $\mathfrak{sl}_2 \cong \mathbb{C}\langle F_i, H_i, E_i\rangle$. Since θ is of maximal heith and $\mathfrak{g}$ is simple, we have $\mathfrak{g} = \mathcal{U}(\mathfrak{g})E_{-\theta} = \mathcal{U}(\mathfrak{n}_+)E_{-\theta}$. Now define $\widetilde{g}_1$ as the Lie algebra generated by $\widetilde{\mathfrak{h}}, e_i, f_i$ $(i = 0, \dots, \ell)$. Since $\mathfrak{g}$ is simple, it's generated by the E_i, F_i's and we have $\mathfrak{g} \subset \widetilde{\mathfrak{g}}$. Furthermore we see $\mathcal{U}(\mathfrak{n}_+)e_0 = \mathfrak{g} \otimes t \in \widetilde{\mathfrak{g}}_1$, and an inudction show that $\mathfrak{g} \otimes t^j \in \widetilde{\mathfrak{g}}_1,\ \forall j \geq 0$. In a similar way we see that $\mathfrak{g} \otimes t^j \in \widetilde{\mathfrak{g}}_1,\ j < 0$. So we conclude that $\widetilde{\mathfrak{g}}_1 = \widetilde{\mathfrak{g}}$.

– ($\widetilde{\mathfrak{g}}$ generators satisfies all the relations).

$$\begin{aligned}
&[e_0, f_0] = -\theta^\vee \otimes 1 + \frac{2}{(\theta,\theta)}K = \alpha_0^\vee \\
&[e_0, f_i] = [E_{-\theta} \otimes t, F_i \otimes 1] = [E_{-\theta}, F_i] \otimes t = 0 \text{ (ht}(\theta) \text{ is minimum)} \\
&[e_0, h] = [E_{-\theta}, h] \otimes t = \theta(h)e_0 = -\langle \delta - \theta, h\rangle e_0 \ (\forall h \in \mathfrak{h}) \\
&[e_0, d] = -e_0 = -\langle \delta - \theta, d\rangle e_0 \\
&[e_0, K] = 0 = -\langle \delta - \theta, K\rangle e_0 \\
&\text{Etc} \dots
\end{aligned} \tag{1.64}$$

– (Any nonzero ideal intersects $\widetilde{h}$ trivially). Let $I \subset \widetilde{\mathfrak{g}}$ be a nonzero ideal. Since $I = \bigoplus_{\alpha+j\delta\neq 0} I \cap \widetilde{\mathfrak{g}}_{\alpha+j\delta}$ we know that there exist $x \otimes t^j \in I \cap \mathfrak{g}_{\alpha+j\delta} \setminus 0$ because of equation (1.58). Since the Killing form is nondegenerate, there exist $y \in \mathfrak{g}_{-\alpha}$

such that $(x|y) \neq 0$. Therefore

$$[y \otimes t^{-j}, x \otimes t^j] = [x,y] \otimes 1 - j(x|y)K \in \widetilde{\mathfrak{h}} \Rightarrow j = 0 \tag{1.65}$$

and $[x,y] = (x|y)\nu^{-1}(\alpha) = 0 \Rightarrow \alpha = 0$. This constradicts the hypothesis $\alpha + j\delta \neq 0$. □

- (Invariant bilinear form). By the previous theorem we know that $\widetilde{g}$ is a Kac - Moody algebra, so it has an unique invariant bilinear $(\cdot|\cdot)$ form satisfying equation (1.22). On Cartan $\widetilde{\mathfrak{h}}$ we must have

$$\begin{aligned}
&(d|d) = 0 \\
&(d|\alpha_i^\vee) = \langle \alpha_i, d\rangle = 0 \\
&(\tfrac{2}{(\theta,\theta)}K - \theta^\vee \otimes 1|\tfrac{2}{(\theta,\theta)}K - \theta^\vee \otimes 1) = 2 \\
&(\tfrac{2}{(\theta,\theta)}K - \theta^\vee \otimes 1|d) = \epsilon_0\langle \delta - \theta, d\rangle = \epsilon_0 \\
&(\tfrac{2}{(\theta,\theta)}K - \theta^\vee \otimes 1|\alpha_i^\vee) = -\epsilon_0\langle \theta, H_i\rangle \ (i > 0)
\end{aligned} \tag{1.66}$$

However, we know that $\mathfrak{h} = \mathrm{Span}(\{H_1, \ldots, H_\ell\})$ and $\langle \alpha_i, \mathbb{C}K + \mathbb{C}d\rangle = 0,\ i > 1$. Therefore, we have $(\mathfrak{h} \otimes 1|\mathbb{C}K + \mathbb{C}d) = 0$. From the previous equation we see that

$$\begin{aligned}
&(\tfrac{2}{(\theta,\theta)}K|\tfrac{2}{(\theta,\theta)}K) = 0 \Rightarrow (K|K) = 0 \\
&(\tfrac{2}{(\theta,\theta)}K|d) = \epsilon_0 \Rightarrow (K|d) = \epsilon_0\frac{(\theta,\theta)}{2}
\end{aligned} \tag{1.67}$$

Remark 1.1.13. To avoid circular arguments, observe that $(\cdot|\cdot)|_{\mathfrak{g}}$ is an invariant bilinear form. So choose a normalization such that $(\theta|\theta) = (\theta, \theta)$, where

$$(\cdot,\cdot) = \frac{1}{2h^\vee}K(\cdot,\cdot) \ (h^\vee \text{ is the dual coxeter number}). \tag{1.68}$$

By equation (1.66), we also have

$$\begin{aligned}
&\frac{2}{(\theta|\theta)}(K|\alpha_i^\vee) - (\theta^\vee \otimes 1|\alpha_i^\vee) = -\epsilon_0\langle \theta, H_i\rangle \\
\iff\ & \frac{2\epsilon_i}{(\theta|\theta)}\underbrace{\langle \alpha_i, K\rangle}_{=0} - (\theta^\vee \otimes 1|\alpha_i^\vee) = -\epsilon_0\langle \theta, H_i\rangle \\
\iff\ & \epsilon_0\langle \theta, H_i\rangle = (\theta^\vee \otimes 1|H_i \otimes 1) \\
\iff\ & \epsilon_0\langle \theta, H_i\rangle = \frac{2}{(\theta|\theta)}(\nu^{-1}(\theta)|H_i) \\
\iff\ & \epsilon_0\langle \theta, H_i\rangle = \frac{2}{(\theta|\theta)}\langle \theta, H_i\rangle \\
\iff\ & \epsilon_0 = \frac{2}{(\theta|\theta)}.
\end{aligned} \tag{1.69}$$

From the previous remark we have $(\theta|\theta) = (\theta, \theta)$ and $\epsilon_0 = 1$. Maybe we can avoid all this calculation by simply choosing $\epsilon_0 = 1$, because the symmetrized form of a GCM can be modified by multiplying the symmetric part by a scalar matrix.

Then we see that $(K|d) = 1$ and by definition we have $\nu(K) = \delta$. We also define $\Lambda_0 = \nu(d)$, so we have

$$\begin{aligned} &(\delta|\delta) = (d|d) = 0 \\ &(\Lambda_0|\Lambda_0) = (K|K) = 0 \\ &(\Lambda_0|\delta) = (K|d) = 1 \end{aligned} \tag{1.70}$$

Now we determine $(\cdot|\cdot)$ on $\widetilde{g}$. It's sufficient to determine the pairing where it's different from zero. So we calculate $(x \otimes t^j | y \otimes t^{-j}), (x, y) \in \mathfrak{g}_\alpha \times \mathfrak{g}_{-\alpha}$:

$$\begin{aligned} \langle \alpha, H_i \rangle (x|y) &= ([H_i, x]|y) \\ &= (H_i \otimes 1 | [x, y] \otimes 1) \\ &= ([H_i, x] \otimes t^j | y \otimes t^{-j}) \\ &= \langle \alpha, H_i \rangle (x \otimes t^j | y \otimes t^{-j}) \end{aligned} \tag{1.71}$$

If $\alpha \neq 0$, then $\alpha(H_i) \neq 0$ for some i and we are done. So we conclude that

$$(x \otimes t^m | y \otimes t^n) = \delta_{m,-n}(x|y). \tag{1.72}$$

This is compatible with the general expression:

$$(x \otimes P | y \otimes Q) = \mathrm{Res}_t(t^{-1}PQ)(x|y). \tag{1.73}$$

- (The Weyl group). First lets compute the Weyl group of $\widetilde{sl}_2$. In this case we have

$\mathfrak{sl}_2 = \mathbb{C}F + \mathbb{C}H + \mathbb{C}E$
$\alpha = \theta : H \mapsto 2$
$\theta^\vee = H$
$\Pi^\vee = \{K - H \otimes 1, H \otimes 1\}$
$\Pi = \{\delta - \alpha, \alpha\}$

Remember the definition of the Weyl group acting on roots (1.1.8).

$$\begin{aligned} r_0 &= \begin{cases} \alpha \mapsto \alpha - \langle \alpha, K - H \otimes 1 \rangle (\delta - \alpha) = 2\delta - \alpha \\ \Lambda_0 \mapsto \Lambda_0 - \langle \Lambda_0, K - H \otimes 1 \rangle (\delta - \alpha) = \Lambda_0 - \delta + \alpha \\ \delta \mapsto \delta - \langle \delta, K - H \otimes 1 \rangle (\delta - \alpha) = \delta \end{cases} \\ &\Rightarrow r_0(m\alpha + k\Lambda_0 + f\delta) = (k - m)\alpha + k\Lambda_0 + (\delta - k + 2m)\delta \end{aligned} \tag{1.74}$$

We see that in the basis $\{\alpha, \Lambda_0, \delta\}$ we have

$$r_1 = \mathrm{Diag}(-1, 1, 1) \text{ and } r_0 = \begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 2 & -1 & 1 \end{pmatrix} \tag{1.75}$$

The Weyl group of $\widetilde{sl}_2$ is $W = \langle r_0, r_1 \rangle$.

Now lets describe the structure of Weyl groups of affine Lie algebras in general. We will write $\widetilde{\lambda} = \lambda + n\delta + k\Lambda_0 \in \widetilde{\mathfrak{h}}^*$, where $\lambda = \mathrm{Proj}(\widetilde{\lambda}) \in \mathfrak{h}^*$. From definition (1.1.8), we see that

$$\begin{cases} r_0(\widetilde{\lambda}) = \widetilde{\lambda} - \dfrac{2}{(\theta|\theta)}(k - (\theta|\lambda))(\delta - \theta) \text{ (because } \langle \delta - \theta, \widetilde{\lambda} \rangle = k - (\theta|\lambda)) \\ r_i(\widetilde{\lambda}) = \widetilde{\lambda} - 2\dfrac{(\lambda|\alpha_i)}{(\alpha_i|\alpha_i)}\alpha_i \end{cases} \tag{1.76}$$

becasue $(\widetilde{\lambda}|h) = (\lambda|h)$, $\forall h \in \mathfrak{h}$. Furthermore, we *define*

$$\begin{aligned} r_\theta(\widetilde{\lambda}) &= \widetilde{\lambda} - 2\frac{(\widetilde{\lambda}|\theta)}{(\theta|\theta)}\theta \\ &= \widetilde{\lambda} - 2\frac{(\lambda|\theta)}{(\theta|\theta)}\theta. \end{aligned} \tag{1.77}$$

Now lets compute

$$\begin{aligned} r_0 r_\theta(\widetilde{\lambda}) &= r_0(\widetilde{\lambda} - 2\frac{(\lambda|\theta)}{(\theta|\theta)}\theta) \\ &= \widetilde{\lambda} - 2\frac{(\lambda|\theta)}{(\theta|\theta)}\theta - \frac{2}{(\theta|\theta)}(k - (\theta|\lambda) + 2\frac{(\theta|\lambda)(\theta|\theta)}{(\theta|\theta)})(\delta - \theta) \\ &= \widetilde{\lambda} + k\frac{2\theta}{(\theta|\theta)} - (\langle \lambda, \theta^\vee \rangle + k\frac{\|\theta^\vee\|^2}{2})\delta \\ &= \widetilde{\lambda} + k\nu(\theta^\vee) - (\langle \lambda, \theta^\vee \rangle + k\frac{\|\theta^\vee\|^2}{2})\delta. \end{aligned} \tag{1.78}$$

This motivates the following notation:

Notation 1.1.14. Let $M = \mathbb{Z}W\nu(\theta^\vee)$, we define the *translation*

$$t_\alpha(\widetilde{\lambda}) = \widetilde{\lambda} + k\alpha - ((\lambda|\alpha) + k\frac{\|\alpha\|^2}{2})\delta, \ \alpha \in M. \tag{1.79}$$

Observe that if we write $\alpha = \nu(\theta^\vee)$, we have $\langle \lambda, \theta^\vee \rangle = (\nu^{-1}(\lambda)|\nu^{-1}(\alpha)) = (\lambda|\alpha)$ and $\|\alpha\|^2 = \|\theta^\vee\|^2$, so our notation is consistent ($t_{\nu(\theta^\vee)} = t_\alpha$).

Lemma 1.1.15. For $w \in W$ and $\alpha, \beta \in M$ we have

(1) $w t_\alpha w^{-1} = t_{w(\alpha)}$

(2) $t_\alpha t_\beta = t_{\alpha+\beta}$

Proof: For the first identity we observe that since $w \in W$ (the Weyl group of $\mathfrak{g}$), we have $w(\widetilde{\lambda}) = w(\lambda) + n\delta + k\Lambda_0$. Therefore

$$
\begin{aligned}
wt_\alpha w^{-1}(\widetilde{\lambda}) &= wt_\alpha(w^{-1}(\lambda) + n\delta + k\Lambda_0) \\
&= w(w^{-1}(\lambda) + n\delta + k\Lambda_0 + k\alpha - ((w^{-1}(\alpha)|\alpha) + k\frac{\|\alpha\|^2}{2})\delta \\
&= \lambda + n\delta + k\Lambda_0 + kw(\alpha) - ((\alpha|w(\alpha)) + k\frac{\|w(\alpha)\|^2}{2})\delta \\
&= t_{w(\alpha)}(\widetilde{\lambda}).
\end{aligned}
\tag{1.80}
$$

For the second one we compute:

$$
\begin{aligned}
t_\alpha t_\beta(\widetilde{\lambda}) &= t_\alpha(\widetilde{\lambda} + k\beta - ((\lambda|\beta) + k\frac{\|\beta\|^2}{2})\delta) \\
&= \widetilde{\lambda} + k\beta - ((\lambda|\beta) + k\frac{\|\beta\|^2}{2})\delta + k\alpha - ((\lambda + k\beta|\alpha) + k\frac{\|\alpha\|^2}{2})\delta \\
&= t_{\alpha+\beta}(\widetilde{\lambda}).
\end{aligned}
\tag{1.81}
$$

Theorem 1.1.16. The set $T = \{t_\alpha : \alpha \in M\}$, $M = \mathbb{Z}W\nu(\theta^\vee)$ is a normal subgroup of the Weyl group $\widetilde{W}$ of $\widetilde{\mathfrak{g}}$. Furthermore we have $\widetilde{W} = TW$, $T \cap W = \{1\}$. In one world, we have the equality:

$$
\widetilde{W} = W \rtimes T. \tag{1.82}
$$

Proof: First observe that $t_{\nu(\theta^\vee)} = r_0 r_\theta \in \widetilde{W}$ and since $t_{w\nu(\theta^\vee)} = wt_{\nu(\theta^\vee)}w^{-1} \in \widetilde{W}$, we see that $T \subset \widetilde{W}$. Furthermore, the identity $wt_\alpha w^{-1} = t_{w(\alpha)}$, $\alpha \in M, w \in W$ implies that $W \subset N(T)$ and since $\widetilde{W} = \langle r_\theta, W\rangle = TW$, we see that T is normal in $\widetilde{W}$. Finally, the quality $W \cap T = \{1\}$ follows because $n \in \mathbb{Z} \mapsto t_{n\alpha}$ is injective. So, since W is finite, we can not have $t_\alpha \in W$, $\alpha \neq 0$. □

Example 1.1.17. The last theorem provide us a nice description of the Weyl group of $\widetilde{\mathfrak{sl}}_2$. In this case we have $M = \mathbb{Z}\alpha$ and

$$
\widetilde{W} = \{t_{m\alpha}\}_{m\in\mathbb{Z}} \cup \{t_{m\alpha} \circ r\}_{m\in\mathbb{Z}},\ r(\alpha) = -\alpha. \tag{1.83}
$$

1.1.5 The Kac character formula

Remark 1.1.18. Let M be a weight module of a Kac - Moody algebra, then we write $\lambda > \mu$ if

$$
\lambda - \mu \in \sum_i \mathbb{Z}_{\geq 0}\alpha_i \tag{1.84}
$$

where $\Pi = \{\alpha_1, \dots, \alpha_n\}$.

- (Verma module). Let $\mathfrak{g} = \mathfrak{g}(A)$ be a Kac - Moody algebra and $\lambda \in \mathfrak{h}^*$, we have a $\mathfrak{h} + \mathfrak{n}_+ -$module $\mathbb{C}v_\lambda$ defined by $\mathfrak{n}_+ v_\lambda = 0$ and $h \cdot v_\lambda = \langle \lambda, h \rangle v_\lambda$. We let

$$M(\lambda) = \mathcal{U}(\mathfrak{g}) \otimes_{\mathcal{U}(\mathfrak{h}+\mathfrak{n}_+)} \mathbb{C}v_\lambda. \tag{1.85}$$

The Verma module is a weight representation. In fact, we have

$$M(\lambda) = \bigoplus_{k_1,\dots,k_n \geq 0} M(\lambda)_{\lambda - \Sigma_i k_i \alpha_i}. \tag{1.86}$$

Moreover

$$\dim(M(\lambda)_\mu) = \text{“ number of ways of expressing } \lambda - \mu \text{ as a sum of positive roots”}$$

For example, let $\mathfrak{g} = \mathfrak{sl}_3$. We have positive roots $\{\alpha_1, \alpha_2, \alpha_3\}$ where $\alpha_3 = \alpha_1 + \alpha_2$. So

$$M(\lambda)_{2\alpha_1+\alpha_2} = \mathbb{C}F_1^2 F_2 v_\lambda + \mathbb{C}F_1 F_3 v_\lambda \tag{1.87}$$

So $\dim M(\lambda)_{2\alpha_1+\alpha_2} = 2$, which is the number of ways of expressing $2\alpha_1 + \alpha_2$ as a sum of elements in $\{\alpha_1, \alpha_2, \alpha_3\}$.

Then we see that, in general $\dim M(\lambda)_\mu < \infty$. Moreover, each weight is bounded by λ, so in particular, this means that $\dim(\mathcal{U}(\mathfrak{n}_+)v) < \infty$ for any $v \in M(\lambda)_\mu$.

Since $M(\lambda)$ is a weight representation, we can define the maximal ideal $J(\lambda)$ intersecting $\mathbb{C}v_\lambda$ trivially. The quocient $L(\lambda) = M(\lambda)/J(\lambda)$ is an irreducible module, because any proper ideal of $M(\lambda)$ must intersects $\mathbb{C}v_\lambda$, so it must be contained in $J(\lambda)$. In general, each highest weight module M with highest weight vector v'_λ is a quotient of $M(\lambda)$. To see this, we need to verify that $M(\lambda) \twoheadrightarrow M : uv_\lambda \mapsto uv'_\lambda$ is well defined. (TODO: proof).

Remark 1.1.19. We see that $L(0)$ is the trivial module. In fact, it's clear that $\mathcal{U}(\mathfrak{g})\mathfrak{g}_{-1}v_0 \subset M(0)$ is a proper submodule of codimension 1. In fact, for each j we have $E_i F_j v_0 = 0$, $\forall i$. In particular

$$\mathrm{ch}(L(0)) = 1 \cdot e^0 = 1. \tag{1.88}$$

- (Category $\mathcal{O}$) The Category $\mathcal{O}$ of a Kac - Moody algebra is the full subcategory of $\mathfrak{g}-$modules whose objects M satisfies:

  (1) M is finitely generated as a $\mathcal{U}(\mathfrak{g})-$module

  (2) M is a weight module

  (3) The action of $\mathfrak{n}_+$ is locally nilpotent, which is equivalent to $\dim(\mathcal{U}(\mathfrak{n}_+)v) < \infty$, $\forall v \in M$.

The third conditions is equivalent to $(3')$ which says that there exist a finite collection of weiths $\{\lambda_1, \ldots, \lambda_r\}$ which are upper bounds for the weigths of M, i.e., $M_\mu \neq 0 \Rightarrow \mu < \lambda_i$ for some i. In fact, suppose that $(3')$ is satisfied and let $v \in M_\mu \setminus 0$. Then

$$\#\{(k_1, \ldots, k_n) \in \mathbb{Z}^n_{\geq 0} \text{ such that } M_{\mu + \sum_j k_j \alpha_j} \neq 0\} < \infty \tag{1.89}$$

because $M_{\mu+\sum_j k_j\alpha_j} \neq 0 \Rightarrow \sum_j k_j\alpha_j < \lambda_i - \mu$ for some i. (TODO: proof of the converse).

It's clear that the Verma modules are objects in the Category $\mathcal{O}$. Furthermore, we have $V, U \in \mathrm{Obj}(\mathcal{O}) \Rightarrow V \oplus U, V \otimes U \in \mathrm{Obj}(\mathcal{O})$ and if $U \subset V$ we also have $V/U \in \mathrm{Obj}(\mathcal{O})$. In particular we see that highest weight modules are ojects in this category.

Let $M \in \mathrm{Obj}(\mathcal{O})$, we define a formal series

$$\mathrm{ch}(M) = \sum_{\mu \in \mathfrak{h}^*} \dim(M_\mu) e^\mu. \tag{1.90}$$

Example 1.1.20. Consider $\mathfrak{g} = \mathfrak{sl}_2$ and let $\lambda = k\varpi \in \mathfrak{h}^*, \varpi(H) = 1$. We have $M(\lambda)_{\lambda - 2m\varpi} = \mathbb{C}F^m v_\lambda$ and the fundamental equation:

$$EF^m v_\lambda = m(k - m + 1)F^{m-1} v_\lambda. \tag{1.91}$$

Then we see that if $k > 0$, then

$$M(\lambda) = M(k\varpi) \supset M(-(k+2)\varpi) \supset 0 \tag{1.92}$$

has the property that each factor is irreducible. Let $x = e^{k\varpi}$, then

$$\begin{aligned} \mathrm{ch}(M(\lambda)) &= x^k + x^{k-2} + x^{k-4} + \cdots = \frac{x^k}{1 - x^{-2}} \\ \mathrm{ch}(M(-(k+2)\varpi)) &= \frac{x^{-k-2}}{1 - x^{-2}} \end{aligned} \tag{1.93}$$

Therefore

$$\mathrm{ch}(L(\lambda)) = \frac{x^k - x^{-k-2}}{1 - x^{-2}} = x^k + x^{k-2} + \cdots + x^{-k}. \tag{1.94}$$

Lemma 1.1.21. The character of a Verma module is given as follows:

$$\mathrm{ch}(M(\lambda)) = \frac{e^\lambda}{\prod_{\alpha \in \Delta_+} (1 - e^{-\alpha})^{m_\alpha}}. \tag{1.95}$$

Proof: For each α, write a basis $\mathfrak{g}_\alpha = \langle e_\alpha^{(i)} \rangle$. We know that $M(\lambda)_\mu \neq 0 \iff \lambda - \mu \in Q_+$. In such a case $\dim(M(\lambda)_\mu)$ is equal to the number of ways to write $\lambda - \mu$ as the sum of positive roots. This corresponds to basis elements of $M(\lambda)_\mu$ given by

$$e_{-\alpha_{i_1}}^{r_1} \cdots e_{-\alpha_{i_k}}^{r_k} v_\lambda, \sum_{j=1}^{k} r_j \alpha_{i_j} = \lambda - \mu. \tag{1.96}$$

So, to count the number of these sequences we just need to take the coefficient of $\lambda - \mu$ in the expansion

$$\prod_{\alpha \in \Delta_+} (1 + e^{-\alpha} + e^{-2\alpha} + \cdots)^{m_\alpha}. \tag{1.97}$$

□

- (A filtration).

Definition 1.1.22. Let $M \in \mathrm{Obj}(\mathcal{O})$ and $\mu \in \mathfrak{h}^*$, a filtration of M with respect to μ is a sequence

$$M = M_0 \supset M_1 \supset \cdots M_t = 0 \tag{1.98}$$

such that for each $i \geq 1$, we have one of the two excludent possibilities:

(1) $(M_{i-1}/M_i)_\lambda = 0, \forall \lambda > \mu$

(2) $(M_{i-1}/M_i)_\mu \cong L(\mu)$

Example 1.1.23. For $\mathfrak{g} = \mathfrak{sl}_2$ and $k > 0$ we have

$$M(\lambda) = M(k\varpi) \supset M(-(k+2)\varpi) \supset 0. \tag{1.99}$$

where $M(k\varpi)/M(-(k+2)\varpi) \cong L(k\varpi)$ and $M(-(k+2)\varpi)_\mu = 0\, \forall \mu > k\varpi$.

Lemma 1.1.24. (Existence of filtrations). For each module $M \in \mathrm{Obj}(\mathcal{O})$ and $\lambda \in \mathfrak{h}^*$ we claim that such filtration (1.98) exists. In this case, the number of ocurrence of $L(\lambda)$ in the factors is independent of the filtration. We will write $[M : L(\lambda)]$ to the number of ocurrences $L(\lambda)$ as a factor in any filtration for $\lambda \in \mathfrak{h}^*$.

(TODO: Proof:)

Lemma 1.1.25. Let $M \in \mathrm{Obj}(\mathcal{O})$ and $\lambda \in \mathfrak{h}^*$, we have

$$\dim(M_\mu) = \sum_{\lambda \in \mathfrak{h}^*} [M : L(\lambda)] \dim(L(\lambda)_\mu). \tag{1.100}$$

Proof: Take a filtration satisfying the condition of the definition (1.98). So

$$\begin{aligned}
\dim(M_\mu) &= \sum_{i\geq 1} \dim(M_{i-1})_\mu - \dim(M_i)_\mu \\
&= \sum_{i\geq 1} \dim(M_{i-1}/M_i)_\mu \\
&= \sum_{\lambda>\mu} [M : L(\lambda)] \dim(L(\lambda)_\mu) \\
&= \sum_{\lambda\in\mathfrak{h}^*} [M : L(\lambda)] \dim(L(\lambda)_\mu) \text{ (because } L(\lambda)_\mu = 0,\ \mu \overset{\neq}{>} \lambda)
\end{aligned} \tag{1.101}$$

- (Integrable modules).

> **Definition 1.1.26.** A $\mathfrak{g}-$module V is called integrable if e_i, f_i $(i = 1, \dots, n)$ acts on V as integrable operators.

Notation 1.1.27.

$$\begin{aligned}
&X = \{\lambda \in \mathfrak{h}^* : \langle\lambda, \alpha_i^\vee\rangle \in \mathbb{Z}\ \forall i\} \text{ (integral weights)} \\
&X^+ = \{\lambda \in \mathfrak{h}^* : \langle\lambda, \alpha_i^\vee\rangle \in \mathbb{Z}_{\geq 0}\ \forall i\} \text{ (integral, dominant weights)}
\end{aligned} \tag{1.102}$$

> **Lemma 1.1.28.** $L(\lambda)$ is integrable if and only if λ is dominant and integral.

Proof: The direction ($\Rightarrow$) follows from the representation theory of $\mathfrak{sl}_2$. Let v_λ be a highest weight vector in $L(\lambda)$. Then, since $L(\lambda)$ is integrable, $f_i^r v_\lambda = 0$ for some $r > 0$. It follows that $\mathrm{Span}(f_i^j v_\lambda : j \geq 0)$ is a finite dimensional $\mathfrak{sl}_2 \cong \mathbb{C}\langle f_i, \alpha_i^\vee, e_i\rangle$ module. It highest weight is λ, so $\langle\lambda, \alpha_i^\vee\rangle \in \mathbb{Z}_{\geq 0}$ as a consequence of the representation theory of $\mathfrak{sl}_2\mathbb{C}$. Now lets show the direction ($\Leftarrow$), so assume $\lambda \in X^+$. The fundamental equation (1.91) shows that

$$e_i f_i^{m+1} v_\lambda = 0,\ m = \lambda(\alpha_i^\vee) + 1 \in \mathbb{Z}_{>0}. \tag{1.103}$$

Furthermore $e_j f_i^{m+1} v_\lambda = 0, j \neq i$ while $h f_i^{m+1} v_\lambda = \langle\lambda - (m+1)\alpha_i, h\rangle f_i^{m+1} v_\lambda$. Therefore $\mathcal{U}(\mathfrak{g}) f_i^{m+1} v_\lambda \subset L(\lambda)$ is a proper submodule, so it should be zero. Therefore $f_i^{m+1} v_\lambda = 0$. Now let $u \in \mathcal{U}(\mathfrak{g})$, the equation

$$f_i^N u v_\lambda = (\mathrm{ad}(f_i) + R_{f_i})^N u = \sum_{j=0}^{N} \binom{N}{j} \mathrm{ad}(f_i)^{N-j} u f_i^j v_\lambda = 0 \tag{1.104}$$

if $N \gg 0$ as a consequence of $\mathfrak{g}$ being an integrable module. To finnish the proof we also observe that e_i is locally nilpotent because $L(\lambda) \in \mathrm{Obj}(\mathcal{O})$. $\square$

- (Generalized Casimir). Let $\rho \in \mathfrak{h}^*$ be such that $\langle \rho, \alpha_i^\vee \rangle = 1,\ \forall i$. Also take dual basis $(u_i, u^i) \in \mathfrak{h} \times \mathfrak{h}, (e_\alpha^{(i)}, e_{-\alpha}^{(i)}) \in \mathfrak{g}_\alpha \times \mathfrak{g}_{-\alpha},\ \forall \alpha \in \Delta_+$. We define the generalized Casimir operator
$$\Omega = 2\nu^{-1}(\rho) + \sum_i u_i u^i + \sum_{\alpha \in \Delta_+} \sum_i e_{-\alpha}^{(i)} e_\alpha^{(i)} \tag{1.105}$$

  It's well defined as an endomorphism of an integrable module M. Furthermore it commutes with $\mathcal{U}(\mathfrak{g})$, so on $L(\lambda)$ it acts as a constant by the Schur's lemma. This constant can be easily obtained by applyin Ω on the highest weight vector v_λ:
$$\begin{aligned}\Omega v_\lambda &= 2\langle \lambda, \nu^{-1}(\rho)\rangle + \sum_i \langle \lambda, u_i \rangle \langle \lambda, u^i \rangle \\ &= 2(\rho|\lambda) + \sum_i (\nu^{-1}(\lambda)|u_i)(\nu^{-1}(\lambda)|u^i) \\ &= 2(\rho|\lambda) + (\nu^{-1}(\lambda)|\nu^{-1}(\lambda)) \\ &= ((\lambda+\rho|\lambda+\rho) - (\rho|\rho)) v_\lambda\end{aligned} \tag{1.106}$$

  Example 1.1.29. For $\widetilde{\mathfrak{g}}$ affine, $\widetilde{\rho} = h^\vee \Lambda_0 + \rho$ works, where $h^\vee$ is the dual coxeter number of $\mathfrak{g}$ and $\rho = \frac{1}{2}\sum_{\alpha \in \Delta_+} \alpha$ is the Weyl vector. (TODO: proof).

- (Kac character formula).

- (Jacobi triple product).